Sailing by Paintapu's field

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Abstract

The residual $\varepsilon(m) = \log_2(1+m) - m$ between the floating-point representation's built-in coarse logarithm and the true logarithm cannot be eliminated by any finite correction architecture. We prove this from five independent directions: polynomial approximation (minimax error is strictly positive at every finite degree), affine recurrence (the coordinate change is not in the affine group), treewidth (the binary tiling's treewidth grows without bound), rational closure (commensurability requires $\ln 2$ rational), and Fourier analysis (the derivative jump forces infinitely many modes). In the surreal numbers, all five reduce to one fact: $\varepsilon(m)$ has birthday ω at every interior dyadic rational.

> "On two occasions I have been asked,—'Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?' ... I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question."
>
> —Charles Babbage

> "You best start believing in ghost stories, Miss Turner... you're in one."
>
> —Hector Barbossa

Scientific notation is introduced early on in education. It is a load-bearing representation in any sense of architecture which bears a load. Written in scientific notation, Avogadro's number and the Planck length are equally compact despite being fifty orders of magnitude apart. In nearly all cases we illustrate three facts about this notation in the following order:

1. *Scale is easy.* The two numbers 6.02×10^{23} and 1.616×10^{-35} fit on the same line.
2. *Multiplication is easy.* Checking "about fifty orders of magnitude" is $23 + 35$.
3. *Addition is hard.* Adding Avogadro's number to the Planck length is a mess. Nobody says why.

Students who ask why this must be are given a hoary allegation that the situation is "opposite" in a positional number system. Those who question such an assertion are told this is just how it is. For many, this is first contact with the beautiful inquisitiveness of the Sciences. We ask, if that is just how it is, why must it be *just* like that? Why must there be an irreducible cost to addition in this logarithmic system?

Within the realm of the Reals, I cannot tell you the answer now; the voyage changes you into someone who cannot be told otherwise. In the Surreal numbers, however, the answer is simple. It must be *just* like that; there must be an irreducible cost to addition because for all interior $m = k/2^p$, $0 < k < 2^p$, the surreal birthday of $\varepsilon(m)$ is ω. No finite birthday suffices. We state it now and prove it as we progress with similar proofs and yarns over the Reals.

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Good sea stories and good proofs share one quality: you don't need to believe for them to be true, but it helps. If you have been lost in a tree and found your way back, you know that horizontal movement is expensive. If you have followed one irrationality proof closely enough to remember the moment it snapped shut, you know what "never" means in mathematics. If you have felt two clocks disagree—music against a metronome, jet lag, any moment where time is out of phase—you know lineal correction is not as easy as it seems.

§ 0. The poor man's logarithm

Scientific notation in base 10 writes $x = a \times 10^E$ with $1 \le a < 10$, but a base is a choice. In base 2, we may write $x = (1+m) \times 2^E$ with $m \in [0,1)$. The exponent E is an integer. The mantissa m is a binary fraction. Everything that follows takes place in base 2.

A floating-point number x in the binade $[2^E, 2^{E+1})$ is stored as an exponent E and a mantissa $m \in [0,1)$, so that $x = 2^E \cdot (1+m)$. The bit pattern encodes E and m separately. If we read the entire bit pattern as an integer, up to bias and scaling the result is $E+m$. Mitchell (1962) observed that $E+m$ is a coarse logarithm of x. The representation contains its own log table.

Velvel Kahan exploited this. To compute a square root, take the logarithm, divide by two, take the antilogarithm. With a slide rule this requires three table lookups. In base-2 floating point the first and third are free—reinterpret the bit pattern as an integer, operate, reinterpret back—and the second is a bit shift. Kahan's "magic square root" is the slide rule method executed on the representation itself, with no table in memory. The representation is the table.

This is not an approximation that happens to work. Day (2023) proves that $L(x) = E + m$ is the unique affine function on the binade that agrees with $\log_2(x)$ at both endpoints. It is the optimal coarse surrogate in a precise sense: the endpoint-exact affine map that minimizes the worst-case deviation from $\log_2$ on $[2^E, 2^{E+1})$.

The deviation has a name. Define

$$\varepsilon(m) = \log_2(1+m) - m, \qquad m \in [0,1). \tag{1}$$

Then $\log_2(x) = L(x) + \varepsilon(m_x)$. The function ε is zero at $m = 0$ and $m \to 1$, positive on $(0,1)$, concave, with a unique maximum at $m^* = 1/\ln 2 - 1 \approx 0.4427$, where $\varepsilon(m^*) \approx 0.0861$. It is determined entirely by $\ln 2$.

```
i = 0x5f3759df - ( i >> 1 ); // inarticulate sailor cursing
```

The constant `0x5f3759df` computes $x^{-1/2}$ by the method of log tables: reinterpret the bit pattern as an integer (the log is free), shift right and subtract (divide by -2 in log-space), reinterpret back (antilog). One Newton iteration refines. The constant encodes the exponent bias (127), the mantissa width (23), and one real parameter: the optimal intercept σ, a functional of ε. Its value depends on the shape of the bump—its endpoints, its maximum, its concavity.

One number suffices because the bump is the same on every binade. The binary subdivision of $[0,1]$ to depth d creates 2^d cells. In the Poincaré half-plane, depth d lives at height $y = 2^{-d}$. Each cell has Euclidean width $1/2^d$ and horocyclic arc length 1, independent of d. Consecutive depths are separated by hyperbolic distance $\ln 2$. Every cell is congruent: horocyclic width 1, geodesic height $\ln 2$. The function ε lives on this cell and does not change from binade to binade: same shape, same peak, same zeros. The optimal intercept is the same on every cell because there is only one cell shape.

Day's decoupling theorem: the coarse stage (computing L) and the correction stage (approximating the residual) are independent problems. The coarse solution is inherited by every correction architecture. The correction stage approximates $z^{-1/b}$ on an interval $[z_{\min}, z_{\max}]$, where the extrema of z are values of $2^{\varepsilon(r/k)}$ at Farey sample points—the mediants of the Stern–Brocot tree. Day's extremal classification is optimization in ε-coordinates.

ε is our object. Everything after this is about what happens when you try to make it vanish with finite means.

§ 1. Polynomial packing on a tile

The congruence of the cell—the prototile—let a single integer absorb the bounds of ε. The coarse stage is free. The natural next move is to improve on it: approximate the correction $z^{-1/b}$ with a polynomial, driving the error toward zero.

Grant the machine an unbounded real-valued accumulator. It reads the bit string $b_1, b_2, \ldots, b_d$ with positional weights $\delta_j = 2^{-j}$ and computes $L(x) = E + m_x$ exactly. No finite-configuration aliasing argument applies: the accumulator can produce 2^d distinct outputs despite having one control state.

The correction stage remains. Having computed $L(x)$ exactly, form the coarse-stage variable $z(x) = x^a \cdot y(x)^b$, where y is the coarse approximation to $x^{-a/b}$. Day shows that z is a bounded continuous periodic function of $L(x)$ with period b, and its range is a closed interval $[z_{\min}, z_{\max}]$. Write $\rho = z_{\max}/z_{\min}$. The correction seeks to approximate $z^{-1/b}$ on this interval. If one restricts to a polynomial corrector $p(z)$ of degree at most n, the minimax relative error is

$$\varepsilon_n^*(\rho) = \min_{\deg p \le n} \max_{z \in [z_{\min},\, z_{\max}]} \frac{\left| z^{-1/b} - p(z) \right|}{z^{-1/b}}. \tag{2}$$

For every finite n and every nondegenerate interval, this error is strictly positive. The function $z^{-1/b}$ is not a polynomial on any interval of positive length. The polynomial wall stands: $\varepsilon_n^*(\rho) > 0$ for every finite n.

To drive ρ toward 1, sample ε at denser Farey fractions. The Farey fractions at depth k are the mediants of the Stern–Brocot tree at level k. Each new depth refines the sampling of ε and tightens the interval $[z_{\min}, z_{\max}]$. For the corrections to close—for finitely many correction terms to determine all subsequent ones—the corrections must satisfy a finite linear recurrence. Equivalently, their generating function must be rational. This is the condition Padé approximation would recover: a rational generating function from finitely many terms.

A finite linear recurrence requires a finite closure of the binary/Farey mismatch: the grids must re-align after finitely many levels. Re-alignment is commensurability. Commensurability requires $\ln 2$ rational. It is not. The corrections do not repeat. No finite collection of correction terms determines the rest.

Each correction is a local computation—optimal within its tile, at its precision. But even this local optimality is fragile. The density of a tile—the thing that makes "optimal on this tile" meaningful—is not a stable object. Bowen (2002, §2.3.1) shows that drilling an arbitrarily small hole in a tile turns the tiling into a packing whose density is ill-defined.

Surreal §1. A surreal number x born on day d has sign expansion $s\colon d \to \{+, -\}$. Day $d < \omega$ implies x is a dyadic rational. Day ω implies the sign expansion is infinite.

§ 2. Affine recurrence

The polynomial wall defeated a static approach: fix a degree, approximate, accept a nonzero residual. The basis was not rich enough no matter the degree. A different strategy: instead of adding polynomial coefficients, track the Fourier modes of ε directly and average the correction into convergence.

The Baumslag–Solitar group $\mathrm{BS}(1,2)$ has two generators. The generator a acts by multiplication: $x \mapsto 2x$. The generator b acts by addition: $x \mapsto x + 1$. The Cayley graph of $\mathrm{BS}(1,2)$ is the binary tiling. Every computation that combines additions and multiplications traces a path

on this graph. Multiplication moves along geodesics. Addition moves along horocycles. The machine adds numbers. Each addition is one tick of additive time.

Under multiplication, mantissa distributions converge exponentially to the logarithmic distribution. The geodesic direction and the mantissa coordinate are matched: multiplication is the natural operation for mantissas. Under addition, they do not converge (Schatte, Lemma 1). At each step, the mantissa of the running sum rotates—the phase of the distribution shifts by approximately $1/n$ at step n, since $\log(n+1)-\log(n) \approx 1/n$. Early steps rotate fast. Late steps rotate slowly. The rotation never stops.

Try to average the rotation away. Define k-fold Cesàro means: $\mathfrak{A}_0(a_n) = a_n$ and $\mathfrak{A}_{k+1}(a_n) = (1/n)\sum_{m=1}^{n} \mathfrak{A}_k(a_m)$. By Schatte's Lemma 6, the leading term of $\mathfrak{A}_k(h_n(r))$ at frequency r has modulus $|2\pi ir + 1|^{-k}$ and argument $2\pi r \log(|a|n) + k \cdot \arg(2\pi ir + 1)^{-1}$. The modulus shrinks with k. The argument still rotates. The phase $2\pi r \log(|a|n)$ increases without bound. For every k, there exist arbitrarily large n_1, n_2 with phases separated by at least $\pi/2$. The separation in sup norm is at least $c \cdot |2\pi i + 1|^{-k}$ for an absolute constant $c > 0$. The k-th Cesàro means are not Cauchy.

Riesz logarithmic means converge (Schatte, Theorem 5). Define $\mathfrak{L}(a_n) = (1/\log n)\sum_{j=1}^{n} a_j/j$. Schatte proves $|\mathfrak{L}(M_n(x)) - \log x| \leq A_9(\sigma^2/a^2 + 1)/\sqrt[3]{\log n}$, uniformly in $1 \leq x \leq e$. The bound vanishes as $n \to \infty$. The Riesz method converges with no residual.

The operative difference is the weighting. Cesàro weights each term uniformly by $1/n$. Riesz weights the j-th term by $1/j$. The weight $1/j$ matches the deceleration of the rotation: at step j, the phase increment is approximately $1/j$, so the weight gives each step influence proportional to how much new phase information it carries. The Riesz weight $1/j$ is the Jacobian of the coordinate change from additive time to multiplicative time. In multiplicative time—$\log 1, \log 2, \log 3, \ldots$—the rotation is approximately uniform. Averaging uniformly in log-time works, and averaging uniformly in log-time is averaging with weight $1/j$ in linear time.

To cross from Cesàro to Riesz, the machine must change its weighting from uniform to logarithmic. The machine ticks in additive time. The mantissa lives in multiplicative time. The coordinate map between the two clocks is $\psi(m) = \log_2(1+m)$. Its deviation from the identity clock is ε.

The machine's operations on the mantissa are affine: additions translate by a constant, multiplications scale by a constant. These generate the group $\mathrm{Aff}^+(\mathbb{R})$ of maps $x \mapsto ax + b$ with $a > 0$. The coordinate change $\psi(m) = \log_2(1+m)$ is not affine—it is logarithmic. The composition of affine maps is affine: $\mathrm{Aff}^+(\mathbb{R})$ is a group. No finite composition of the machine's operations produces ψ.

Surreal §2. Walker clock: $t \in \mathbb{N}$, one bit per tick. Tree clock: $\tau = \log_2(1+m)$, $m \in [0,1)$. Drift: $\varepsilon(m) = \tau - m$.

§ 3. Treewidth crawl

The polynomial wall and the affine recurrence wall were both defeated by the structure of the problem—one by the spectral shape of ε, the other by the clock mismatch between additive and multiplicative time. Both used the binary tiling as background. Now we use it as the instrument.

The congruent cell from the previous sections—horocyclic width 1, geodesic height $\ln 2$—embeds in the Poincaré half-plane as a tile. This is Bowen's stripe model (2002, §1.2.2) with strip width $W = \ln 2$. The binary subdivision of $[0,1]$ to depth d produces 2^d such tiles. There are uncountably many combinatorially different binary tilings sharing these metric parameters and edge-matching rules. The mantissa grid is the canonical member: the one corresponding to the standard dyadic subdivision.

The binary address of a cell—its d-bit string—is the tail sequence (Dolbilin & Frettlöh, Definition 3.4). All 2^d binary strings are realized at depth d. The number of distinct k-coronae

is 2^{k-1} (Proposition 4.3). The tiling is non-crystallographic: no finite local template captures its global structure (Theorem 4.4).

Kisfaludi-Bak et al. (2023), Proposition 1.3: the treewidth of any n-vertex planar δ-hyperbolic graph is $O(\delta \log n)$. The binary tiling is planar and constant-hyperbolic. A patch of $n = 2^d$ cells at depth d has treewidth $\Theta(d)$.

A finite-state machine with q states reading d bits is a width-q read-once branching program. Babai, Hajnal, Szemerédi, and Turán (1987) give a C^n lower bound on the size of read-once-only branching programs computing an explicit Boolean function. If the correction function falls in the class their techniques reach—and the treewidth growth $\Theta(d)$ of the underlying graph is the natural parameter—then the machine cannot distinguish all local types once d is large enough. It must assign identical corrections to cells whose neighborhoods differ. The displacement field $\Delta^L = -\varepsilon$ varies across those neighborhoods: zero at the binade boundaries, maximal near $m^* \approx 0.4427$, concave between. The residual is nonzero because the field varies where the machine cannot distinguish.

The tiling has one more property worth recording. Every binary tiling has a protrusion direction—a continuous, equivariant map to the ideal boundary. This is a topological invariant, not a choice. Bowen (2002, §2.3.1): the binary tiling space admits no $\mathrm{PSL}(2, \mathbb{R})$-invariant probability measure. The protrusion direction kills every candidate.

The treewidth wall does not close the gap. But the level-by-level construction of $?(x)$—each level absolutely continuous, the limit singular—offers something. If the corrections between levels could be made to close, the singular limit would be reachable from finite data. A glimpse of the path ahead.

Surreal §3. $n = 2^d$ nodes at depth d: $\mathrm{tw} = \Theta(d)$. Walker width: q. $d > f(q)$ implies aliasing.

§ 4. The Padé ghost

We have only a faint hope. We can call on the ghost of the greatest dead numerical analyst Henri Padé. If we get him close enough, maybe his genuinely unearthly approximants can go the distance. Padé approximants are rational functions—ratios of polynomials—fitted to match as many terms of a power series as their degrees allow, reaching past the radius where polynomial truncation fails.

Build $?(x)$ level by level from the Stern–Brocot tree. At each level, deform the metric on each Farey interval to match the corresponding dyadic interval. The level-n floor is absolutely continuous by construction. Integration against it works. The construction gives Padé the best possible perch: rational branching ratios on a tree whose geometry has been aligned at every finite level.

For Padé to close the level-to-level corrections, those corrections must satisfy a finite linear recurrence—equivalently, have a rational generating function. On this scaffold, that would amount to a finite closure of the binary/Farey mismatch: the grids re-align after finitely many levels, and the corrections repeat. The scaffold is built for it.

The grids never re-align. The scaffold's spacing is $\ln 2$, and $\ln 2$ is irrational. No finite closure occurs. The rational generating function that should be there—that the scaffold was built to produce—is not there.

Surreal §4. Corrections closing $\Longrightarrow \{c_k\}$ satisfies a finite linear recurrence $\Longrightarrow \sum c_k z^k$ rational $\Longrightarrow \ln 2 \in \mathbb{Q}$. But $\ln 2 \notin \mathbb{Q}$. $\square$

§ 5. The wall

The polynomial wall's residual at degree n is the projection of $z^{-1/b}$ onto the modes that n coefficients cannot capture. The Hölder wall's residual at order k is the projection of the mantissa oscillation onto the modes that k levels of averaging cannot damp. The treewidth wall's residual is the projection of the displacement field onto the coronas that q states cannot distinguish. Three instruments, one spectral tail.

If ε had finitely many nonzero Fourier modes, all three walls would vanish. The polynomial approximation would close at finite degree. The corrections would satisfy a finite linear recurrence. The Cesàro means would converge. The Padé ghost would not be a ghost.

The Fourier coefficients of ε on the binade circle decay as $O(1/n^2)$ but never terminate. They never terminate because the periodic extension of ε is continuous but not C^1 at the glued endpoints: $\varepsilon'(0^+) = 1/\ln 2 - 1 \approx 0.443$, while $\varepsilon'(1^-) = 1/(2\ln 2) - 1 \approx -0.279$. A finite trigonometric polynomial is C^∞. The derivative jump forces the series to be infinite. The $O(1/n^2)$ decay is the standard consequence of a function that is continuous but not C^1.

The wall is nonzero at every interior point for a separate reason: ε is transcendental there. At every interior machine number $m = k/2^p$, the residual $\varepsilon(k/2^p) = \log_2(1 + k/2^p) - k/2^p$ is transcendental. If $\beta = \log_2(1 + m)$ were algebraic and irrational, then $2^\beta = 1 + m$ algebraic contradicts Gelfond–Schneider. If β were rational, say p/q, then $1+m = 2^{p/q}$, which is impossible for interior dyadics. So β is transcendental, and $\varepsilon = \beta - m$ is transcendental. The correction is never exact.

How close does ε come to zero? Baker: $|b_1 \log \alpha_1 + b_2 \log \alpha_2| \geq \exp\{-C(\log p)^2\}$ for algebraic α_i and integer b_i with $|b_i| \leq p$. At precision p, the clearance from any rounding boundary is at least $2^{-O(p^2)}$. The guard bit budget grows quadratically with precision and depends on no information from the previous precision.

Surreal §5. At each non-root node, three operations: descend left (append $-$), descend right (append $+$), ascend (delete last sign). No horizontal primitive exists. To reach a node at the same depth on a different branch, ascend to common ancestor and re-descend. Cost of horizontal movement: at least 2, at most $2d$.

§ 6. The fire

Here we are. Desperate for any help. We will believe anything, give anything, pay any cost to resolve this. In our moment of despair, we are met by a demon with an offer. It knows of a game we only need to win once, somewhere in the universe. One last game, dyadic on the binary tree. Martin's theorem tells us all Borel games are determined. The result is proved by showing that one player has a winning strategy against every possible strategy of the other.

Player II is the singular structure of $\mu_?$. If we face it alone, we already know we have lost. But this demon is special, it says. Not Descartes' demon—that one asks whether you can trust what you know, and the answer is philosophical. Not Maxwell's demon, though this also sits at a boundary between two regimes. Shannon's demon. It observes one tile at a time. It sorts.

The demon wins if it can do two things, chained. First: establish an information asymmetry across the boundary between the absolutely continuous class and the singular class. Second: aggregate what it learns. Without asymmetry there is nothing to exploit. With asymmetry but without aggregation, the demon knows something it cannot use.

Phase 1. The demon generates a smooth hallucination $\tilde{\varepsilon}$: a function on $[0, 1]$ with the right general shape, the right symmetry under $m \mapsto 1 - m$, the right first few Fourier modes. The hallucination is absolutely continuous. It lives on the smooth continent. Its generation cost is $O(1)$—independent of precision, independent of the depth of the tree.

Phase 2. The demon feeds $\tilde{\varepsilon}$ into a process that interacts with ε. Not a comparison—the demon does not ask "how far is $\tilde{\varepsilon}$ from ε?" That question costs as much as computing ε. Instead,

the process performs its own work: rounding, correction, evaluation—whatever engages with $\log_2$ at the points where it must. The process responds. Its response depends on ε. The cost of the response is borne by the process, not by the demon.

Phase 3. The demon reads the response. Not the full computation—a partial, noisy, stochastic reading. On which side of a rounding boundary did the evaluation fall? Which narrow passage was threaded, and in which direction? One bit, a few bits, recovered at cost $O(n)$—polynomial in the depth, not exponential, not proportional to the cost of recomputation. The demon is eavesdropping, not recomputing.

The first condition is met. ε is transcendental at every interior machine number. The clearance is never zero. Information about the gap exists at every precision (Baker). The asymmetry is real. The demon clears Phase 1.

The second condition is not met. Whatever the demon recovers, it must aggregate. Local observations—per-binade, per-tile—must compose into a global correction. Bowen (2002, §2.1, eq. 2.5) proves—using the Benjamini–Schramm mass-transport principle—that this conversion has exactly one legitimate form: integration of local density ratios against an invariant measure on the tiling space. No other aggregation procedure is available that respects the isometry group.

The invariant measure is the only mediator. We already know it does not exist. The protrusion direction killed it. The per-tile density that seemed obvious was never stable. The demon's three phases are unconstrained. Its strategy is not defeated by a resource bound, a complexity barrier, or a clever counterargument. It is defeated because the object it must eventually produce—the invariant measure that would let local readings compose into a global correction—is not there.

The gap between what the representation gives and what the operations require is ε. It is transcendental at every point where it is nonzero. It lives on a tiling whose invariant measure does not exist. It is the reason addition is hard.

Surreal §6.

Let $m = k/2^p$, $0 < k < 2^p$.

(i) Suppose $\varepsilon(m) = \log_2(1 + k/2^p) - k/2^p$ is a dyadic rational $a/2^r$. Then $\log_2(1 + k/2^p) = k/2^p + a/2^r = s/t$ for some rational s/t in lowest terms. Then $2^{s/t} = 1 + k/2^p$. For $t = 1$: $2^s = 1 + k/2^p$, but $0 < k < 2^p$ puts the right-hand side in $(1, 2)$, so no integer s works. For $t \geq 2$: $2^{s/t}$ is irrational (if $2^{s/t} = a/b$ then $a^t = 2^s b^t$, forcing $b = 1$ and $t \mid s$, contradicting $\gcd(s, t) = 1$), but $1 + k/2^p$ is rational. Therefore $\varepsilon(m)$ is not a dyadic rational.

(ii) A surreal number of finite birthday is a dyadic rational. $\varepsilon(m)$ is not a dyadic rational. Therefore the birthday of $\varepsilon(m) \geq \omega$. Every real number is born by day ω, and $\varepsilon(m)$ is real. Therefore the birthday of $\varepsilon(m) = \omega$.

(iii) For finitely many corrections to determine all subsequent ones, the corrections must satisfy a finite linear recurrence. By §4, this requires $\ln 2 \in \mathbb{Q}$. Therefore no finite collection of corrections at one set of nodes determines the corrections at the rest.

(iv) Every surreal number of finite birthday is a dyadic rational. The target $\varepsilon(m)$ has birthday ω (ii). The corrections do not recur (iii). Therefore closing the gap requires the full ω-stage of the surreal construction, and no finite collection of corrections determines the rest.

The birthday gap cannot be closed by finite means. □

§ 7. Coda

The logarithm is free. Its residual is not. The residual is the price of using an additive grid to index a multiplicative world, and that price is controlled by the Diophantine properties of e and $\ln 2$—two numbers that, if Schanuel's conjecture holds, are algebraically independent: no

nontrivial polynomial with integer coefficients vanishes at $(e, \ln 2)$, at any scale, at any precision, down to the foam.

To get there we had to crawl a firewall we may not cross, even in our imaginations. I hope you know that now. I hope the voyage changes who you are. **That** is the beautiful inquisitiveness of the Sciences.

Surreal §7. Project the tree onto a tape. Descend left: write $-$, advance. Descend right: write $+$, advance. Ascend: retract. The walker is a Turing machine. The sign expansion is the tape.

Every door the paper closed—polynomial, affine, spectral, rational, geometric—was closed because we could name the obstruction and prove it present. The door left open is the finite-state machine: whether a q-state walker can compute the correction at all depths. That door stays open because closing it for all q would require deciding what bounded machines can compute. We choose a base. We compute to a precision. The machine cannot stop for us, probably.

This machine has been built. Gosper (1972) represents the pending arithmetic of two continued fractions as eight integer variables and a rule: consume an input term or emit an output term, each a linear map on the state. The transducer walks the Stern–Brocot tree—our tree—consuming and emitting terms one at a time. It is the most powerful finite walker ever constructed for exact arithmetic on this tree. Its state grows without bound on non-Hurwitz inputs: the treewidth wall. It reaches periodicity only when the continued fraction terms are eventually polynomial: the recurrence wall. Its operations are Möbius transformations that cannot produce the logarithmic coordinate change: the affine wall. Gosper's machine puts in the right figures and gets the right answers, term by term, for as long as you let it run. It just never finishes.

One thread is left hanging.

The treewidth wall (§3) claims that a width-q branching program cannot distinguish all local types on a graph of treewidth $\Theta(d)$. The treewidth bound is a theorem (Kisfaludi-Bak et al., Proposition 1.3). The qualitative consequence—that a fixed-width machine must eventually alias—follows from the treewidth growing without bound while the machine's width stays fixed. But the quantitative bound—the specific relationship between width q and treewidth t that forces aliasing—is conjectured, not proved, for this correction function. We cite Babai, Hajnal, Szemerédi, and Turán for the existence of exponential lower bounds in the read-once branching program model, and we observe that the treewidth of the binary tiling is the natural parameter, but we do not prove that the correction function falls in the class their techniques reach.

We conjecture the bound exists. We do not show it. You are the reader this paper was written for—this is where you can improve on what we have done. Don't let anyone tell you what's easy or hard.

§ 8. Acknowledgements

This is for my wife, my mother, and my daughter. This is because of them and everyone else who ever believed in me.

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